

SYDE - 572 Assignment 1.

Q1. $y = x^2 + 5$

$(0, 0) \quad (-4, 0) \quad (-8, 0) \quad (2, 0) \quad (6, 0)$

Analytical Solution

Distance between a point (x_0, y_0) to a function: $d = \sqrt{(x - x_0)^2 + (f(x) - y_0)^2}$

Minimize distance: $\min D(x) = (x - x_0)^2 + (f(x) - y_0)^2$

Derivative $D(x)$: $D'(x) = 2(x - x_0) + 2(f(x) - y_0) f'(x)$

Minimize $D'(x)$: $0 = 2(x - x_0) + 2(f(x) - y_0) f'(x)$

Input function: $f(x) = x^2 + 5 \longrightarrow f'(x) = 2x$

$$0 = 2(x - x_0) + 2(x^2 + 5 - y_0) \cdot 2x$$

$$0 = 2x - 2x_0 + 4x^3 + 20x - 4xy_0$$

$$0 = 4x^3 + 22x - 2x_0 - 4xy_0$$

$$0 = 2x^3 + 11x - x_0 - 2xy_0$$

Input points:

$(0, 0) \quad 0 = 2x^3 + 11x - (0) - 2x(0)$

$$0 = 2x^3 + 11x$$

$$0 = x(2x^2 + 11)$$

$$x = 0 \longrightarrow f(x) = (0)^2 + 5 = 5 \longrightarrow (0, 5)$$

$$d_{\min} = \sqrt{(0-0)^2 + (5-0)^2} = 5$$

$(-4, 0) \quad 0 = 2x^3 + 11x - (-4) - 2x(0)$

$$0 = 2x^3 + 11x + 4$$

$$x = -0.355470 \longrightarrow f(x) = (-0.355470)^2 + 5 = 5.126359 \longrightarrow (-0.355470, 5.126359)$$

$$d_{\min} = \sqrt{(-0.355470 + 4)^2 + (5.126359 - 0)^2} = 6.289845$$

$(-8, 0) \quad 0 = 2x^3 + 11x - (-8) - 2x(0)$

$$0 = 2x^3 + 11x + 8$$

$$x = -0.672078 \longrightarrow f(x) = (-0.672078)^2 + 5 = 5.451689 \longrightarrow (-0.672078, 5.451689)$$

$$d_{\min} = \sqrt{(-0.672078 + 8)^2 + (5.451689 - 0)^2} = 9.133420$$

$(2, 0) \quad 0 = 2x^3 + 11x - 2 - 2x(0)$

$$0 = 2x^3 + 11x - 2$$

$$x = 0.180745 \longrightarrow f(x) = (0.180745)^2 + 5 = 5.032669 \longrightarrow (0.180745, 5.032669)$$

$$d_{\min} = \sqrt{(0.180745 - 2)^2 + (5.032669 - 0)^2} = 5.351396$$

$(6, 0) \quad 0 = 2x^3 + 11x - 6 - 2x(0)$

$$0 = 2x^3 + 11x - 6$$

$$x = 0.519904 \longrightarrow f(x) = (0.519904)^2 + 5 = 5.270300 \longrightarrow (0.519904, 5.270300)$$

$$d_{\min} = \sqrt{(0.519904 - 6)^2 + (5.270300 - 0)^2} = 7.603125$$

Newton-Raphson Method

Test differentiable: $f(x) = x^2 + 5 \rightarrow f'(x) = 2x + 5 \rightarrow f''(x) = 2$

Set-up: $D(x) = (x - x_0)^2 + (f(x) - y_0)^2$

$$D'(x) = 2x^2 + 11x - x_0 - 2xy_0$$

$$D''(x) = 6x + 11 - 2y_0$$

$$x_{n+1} = x_n - \frac{D'(x)}{D''(x)} = x_n - \frac{2x_n^2 + 11x_n - x_0 - 2x_n y_0}{6x_n + 11 - 2y_0}$$

Input point:

$$(0, 0) \quad x_{n+1} = x_n - \frac{2x_n^2 + 11x_n}{6x_n + 11}$$

$$x_0 = 0$$

$$x_1 = 0 - \frac{2(0)^2 + 11(0)}{6(0) + 11} = 0$$

$$x^* = 0 \rightarrow f(x^*) = (0)^2 + 5 = 5 \rightarrow (0, 5)$$

$$d_{\text{min}} = \sqrt{(0-0)^2 + (5-0)^2} = 5$$

$$(-4, 0) \quad x_{n+1} = x_n - \frac{2x_n^2 + 11x_n + 4}{6x_n + 11}$$

$$x_0 = -4$$

$$\textcircled{1} \quad x_1 = -4 - \frac{2(-4)^2 + 11(-4) + 4}{6(-4) + 11} = -2.429907$$

$$\textcircled{2} \quad x_2 = -2.429907 - \frac{2(-2.429907)^2 + 11(-2.429907) + 4}{6(-2.429907) + 11} = -1.322279$$

$$\textcircled{3} \quad x_3 = -1.322279 - \frac{2(-1.322279)^2 + 11(-1.322279) + 4}{6(-1.322279) + 11} = -0.616439$$

$$\vdots$$

$$x^* = -0.355470 \rightarrow f(x^*) = (-0.355470)^2 + 5 = 5.126359 \rightarrow (-0.355470, 5.126359)$$

$$d_{\text{min}} = \sqrt{(-0.355470 + 4)^2 + (5.126359 - 0)^2} = 6.289845$$

$$(-8, 0) \quad x_{n+1} = x_n - \frac{2x_n^2 + 11x_n + 8}{6x_n + 11}$$

$$x_0 = -8$$

$$\textcircled{1} \quad x_1 = -8 - \frac{2(-8)^2 + 11(-8) + 8}{6(-8) + 11} = -5.205063$$

$$\textcircled{2} \quad x_2 = -5.205063 - \frac{2(-5.205063)^2 + 11(-5.205063) + 8}{6(-5.205063) + 11} = -3.296205$$

$$\textcircled{3} \quad x_3 = -3.296205 - \frac{2(-3.296205)^2 + 11(-3.296205) + 8}{6(-3.296205) + 11} = -1.985209$$

$$\vdots$$

$$x^* = -0.762078 \rightarrow f(x^*) = (-0.762078)^2 + 5 = 5.451690 \rightarrow (-0.762078, 5.451690)$$

$$d_{\text{min}} = \sqrt{(-0.762078 + 8)^2 + (5.451690 - 0)^2} = 9.133470$$

$$(2, 0) \quad x_{n+1} = x_n - \frac{2x_n^2 + 11x_n - 2}{6x_n + 11}$$

$$x_0 = 2$$

$$\textcircled{1} \quad x_1 = 2 - \frac{2(2)^2 + 11(2) - 2}{6(2) + 11} = 0.971429$$

$$\textcircled{2} \quad x_2 = 0.971429 - \frac{2(0.971429)^2 + 11(0.971429) - 2}{6(0.971429) + 11} = 0.340105$$

$$\textcircled{3} \quad x_3 = 0.340105 - \frac{2(0.340105)^2 + 11(0.340105) - 2}{6(0.340105) + 11} = 0.180745$$

$$\vdots$$

$$x^* = 0.180745 \rightarrow f(x^*) = (0.180745)^2 + 5 = 5.032669 \rightarrow (0.180745, 5.032669)$$

$$d_{\text{min}} = \sqrt{(0.180745 - 2)^2 + (5.032669 - 0)^2} = 5.351396$$

$$(6, 0) \quad x_{n+1} = x_n - \frac{2x_n^2 + 11x_n - 6}{6x_n + 11}$$

$$x_0 = 6$$

$$\textcircled{1} \quad x_1 = 6 - \frac{2(6)^2 + 11(6) - 6}{6(6) + 11} = 3.832599$$

$$\textcircled{2} \quad x_2 = 3.832599 - \frac{2(3.832599)^2 + 11(3.832599) - 6}{6(3.832599) + 11} = 2.332075$$

$$\textcircled{3} \quad x_3 = 2.332075 - \frac{2(2.332075)^2 + 11(2.332075) - 6}{6(2.332075) + 11} = 1.300270$$

$$\vdots$$

$$x^* = 0.519904 \rightarrow f(x^*) = (0.519904)^2 + 5 = 5.270300 \rightarrow (0.519904, 5.270300)$$

$$d_{\text{min}} = \sqrt{(0.519904 - 6)^2 + (5.270300 - 0)^2} = 7.603175$$

- Golden Section Search

Set up: $f(x) = x^2 + 5$ on $[a, b]$

$$\phi = \frac{1+\sqrt{5}}{2} = 1.618034 \quad \psi = 2 - \phi = 0.381966$$

$$D(x) = (x - x_0)^2 + (f(x) - y_0)^2 = (x - x_0)^2 + (x^2 + 5 - y_0)^2$$

$$x_1 = a + \psi(b - a)$$

$$x_2 = b - \psi(b - a)$$

Input point

$(0, 0) \rightarrow [-1, 1] \quad a = -1 \quad b = 1$

① $x_1 = -1 + 0.381966(1+1) = -0.236068$

$x_2 = 1 - 0.381966(1+1) = 0.236068$

$D(x_1) = (-0.236068 - 0)^2 + ((-0.236068)^2 + 5)^2 = 25.616115$

$D(x_2) = (0.236068 - 0)^2 + ((0.236068)^2 + 5)^2 = 25.616115$

$[-0.236068, 1]$

② $x_1 = 0.236068$

$x_2 = 1 - 0.381966(1 + 0.236068) = 0.527864$

$D(x_1) = 25.616115$

$D(x_2) = (0.527864 - 0)^2 + ((0.527864)^2 + 5)^2 = 28.141685$

$D(x_2) > D(x_1)$

$[-0.236068, 0.527864]$

③ $x_1 = -0.236068 + 0.381966(0.527864 + 0.236068) = 0.055728$

$x_2 = 0.236068$

$D(x_1) = (0.055728 - 0)^2 + ((0.055728)^2 + 5)^2 = 25.034171$

$D(x_2) = 25.616115$

$D(x_2) > D(x_1)$

$[-0.236068, 0.236068]$

$x^* = 0 \rightarrow f(x^*) = (0)^2 + 5 = 5 \rightarrow [0, 5]$

$dist = \sqrt{(0-0)^2 + (5-0)^2} = 5$

$(-4, 0) \rightarrow [-4, 0] \quad a = -4 \quad b = 0$

① $x_1 = -4 + 0.381966(0+4) = -2.472136$

$x_2 = 0 - 0.381966(0+4) = -1.527864$

$D(x_1) = (-2.472136 + 4)^2 + ((-2.472136)^2 + 5 - 0)^2 = 125.798827$

$D(x_2) = (-1.527864 + 4)^2 + ((-1.527864)^2 + 5 - 0)^2 = 59.904418$

$D(x_1) > D(x_2)$

$[-2.472136, 0]$

② $x_1 = -1.527864$

$x_2 = 0 - 0.381966(0 + 2.472136) = -0.944272$

$D(x_1) = 59.904418$

$D(x_2) = (-0.944272 + 4)^2 + ((-0.944272)^2 + 5 - 0)^2 = 44.049007$

$D(x_1) > D(x_2)$

$[-1.527864, 0]$

③ $x_1 = -0.944272$

$x_2 = 0 - 0.381966(0 + 1.527864) = -0.583592$

$D(x_1) = 44.049007$

$D(x_2) = (-0.583592 + 4)^2 + ((-0.583592)^2 + 5 - 0)^2 = 40.193635$

$D(x_1) > D(x_2)$

$[-0.944272, 0]$

$x^* = -0.355470 \rightarrow f(x^*) = (-0.355470)^2 + 5 = 5.126359 \rightarrow [-0.355470, 5.126359]$

$dist = \sqrt{(-0.355470 + 4)^2 + (5.126359 - 0)^2} = 6.289845$

$$(-8, 0) \rightarrow [-8, 0] \quad a = -8, \quad b = 0$$

$$\textcircled{1} \quad x_1 = -8 + 0.381966(0+8) = -4.944272$$

$$x_2 = 0 - 0.381966(0+8) = -3.055728$$

$$D(x_1) = (-4.944272+8)^2 + ((-4.944272)+5-0)^2 = 876.394068$$

$$D(x_2) = (-3.055728+8)^2 + ((-3.055728)+5-0)^2 = 230.008990$$

$$D(x_1) > D(x_2)$$

$$[-4.944272, 0]$$

$$\textcircled{2} \quad x_1 = -3.055728$$

$$x_2 = 0 - 0.381966(0+4.944272) = -1.888544$$

$$D(x_1) = 230.008990$$

$$D(x_2) = (-1.888544+8)^2 + ((-1.888544)+5-0)^2 = 110.736494$$

$$D(x_1) > D(x_2)$$

$$[-3.055728, 0]$$

$$\textcircled{3} \quad x_1 = -1.888544$$

$$x_2 = 0 - 0.381966(0+3.055728) = -1.167184$$

$$D(x_1) = 110.736494$$

$$D(x_2) = (-1.167184+8)^2 + ((-1.167184)+5-0)^2 = 87.166475$$

$$D(x_1) > D(x_2)$$

$$[-1.888544, 0]$$

$$x^* = -0.672078 \rightarrow f(x^*) = (-0.672078)^2 + 5 = 5.451699$$

$$\rightarrow [-0.672078, 5.451699]$$

$$d_{\min} = \sqrt{(-0.672078+8)^2 + (5.451699-0)^2} = 9.133470$$

$$(2, 0) \rightarrow [0, 2] \quad a=0 \quad b=2$$

$$\textcircled{1} \quad x_1 = 0 + 0.381966(2-0) = 0.763932$$

$$x_2 = 2 - 0.381966(2-0) = 1.236068$$

$$D(x_1) = (0.763932-2)^2 + ((0.763932)+5-0)^2 = 32.704365$$

$$D(x_2) = (1.236068-2)^2 + ((1.236068)+5-0)^2 = 43.196601$$

$$D(x_1) < D(x_2)$$

$$[0, 1.236068]$$

$$\textcircled{2} \quad x_1 = 0 + 0.381966(1.236068-0) = 0.472136$$

$$x_2 = 0.763932$$

$$D(x_1) = (0.472136-2)^2 + ((0.472136)+5-0)^2 = 29.613184$$

$$D(x_2) = 32.704365$$

$$D(x_1) < D(x_2)$$

$$[0, 0.763932]$$

$$\textcircled{3} \quad x_1 = 0 + 0.381966(0.763932-0) = 0.291796$$

$$x_2 = 0.472136$$

$$D(x_1) = (0.291796-2)^2 + ((0.291796)+5-0)^2 = 28.776660$$

$$D(x_2) = 29.613182$$

$$D(x_1) > D(x_2)$$

$$[0.472136, 0]$$

$$x^* = 0.180340 \rightarrow f(x^*) = (0.180340)^2 + 5 = 5.032669$$

$$\rightarrow [0.180340, 5.032669]$$

$$d_{\min} = \sqrt{(0.180340-2)^2 + (5.032669-0)^2} = 5.351396$$

$$(b, 0) \rightarrow [0, b] \quad a=0, b=b$$

$$\textcircled{1} \quad x_1 = 0 + 0.381966(b-0) = 2.291796$$

$$x_2 = b - 0.381966(b-0) = 3.708204$$

$$D(x_1) = (2.291796 - b)^2 + ((2.291796)^2 + 5 - 0)^2 = 118.861031$$

$$D(x_2) = (3.708204 - b)^2 + ((3.708204)^2 + 5 - 0)^2 = 356.843945$$

$$D(x_1) < D(x_2)$$

$$[0, 3.708204]$$

$$\textcircled{2} \quad x_1 = 0 + 0.381966(3.708204 - 0) = 1.416408$$

$$x_2 = 2.291796$$

$$D(x_1) = (1.416408 - b)^2 + ((1.416408)^2 + 5 - 0)^2 = 70.096313$$

$$D(x_2) = 118.861031$$

$$D(x_1) < D(x_2)$$

$$[0, 2.291796]$$

$$\textcircled{3} \quad x_1 = 0 + 0.381966(2.291796 - 0) = 0.875388$$

$$x_2 = 1.416408$$

$$D(x_1) = (0.875388 - b)^2 + ((0.875388)^2 + 5 - 0)^2 = 59.511914$$

$$D(x_2) = 70.096313$$

$$D(x_1) < D(x_2)$$

$$[0, 1.416408]$$

$$x^* = 0.519904 \rightarrow f(x^*) = (0.519904)^2 + 5 = 5.270300$$

$$\rightarrow [0.519904, 5.270300]$$

$$d_{\min} = \sqrt{(0.519904 - b)^2 + (5.270300 - 0)^2} = 7.603125$$

Q2. (0, 0.5) (2, 3.5) (1, 1.5) (3, 7.5)

— Line

— Analytical: $y = mx + b$
 $\hat{y}_i = mx_i + b \rightarrow \text{error } e_i = \hat{y}_i - y_i$

For 4 points: $MSE(m, b) = \frac{1}{4} \sum_{i=1}^4 (mx_i + b - y_i)^2$

$$= \frac{1}{4} [(b - 0.5)^2 + (2m + b - 3.5)^2 + (m + b - 1.5)^2 + (3m + b - 7.5)^2]$$

$$= \frac{1}{4} [b^2 - b + 0.25 + 4m^2 + b^2 + 4mb - 14m - 7b + 12.25 + m^2 + b^2 + 2mb - 3m - 3b + 2.25 + 9m^2 + b^2 + 6mb - 45m - 15b + 56.25]$$

$$= \frac{1}{4} [14m^2 + 4b^2 + 12mb - 62m - 26b + 71]$$

$$= 3.5m^2 + b^2 + 3mb - 15.5m - 6.5b + 17.75$$

$$\frac{\partial MSE(m, b)}{\partial m} = 7m + 3b - 15.5 = 0 \rightarrow \begin{cases} 7m + 3b = 15.5 & \textcircled{1} \\ 3m + 2b = 6.5 & \textcircled{2} \end{cases}$$

$$\frac{\partial MSE(m, b)}{\partial b} = 2b + 3m - 6.5 = 0 \rightarrow \begin{cases} 7m + 3b = 15.5 & \textcircled{1} \\ 3m + 2b = 6.5 & \textcircled{2} \end{cases}$$

$$\textcircled{2} \quad b = \frac{6.5 - 3m}{2}$$

$$\textcircled{1} \quad 7m + 3\left(\frac{6.5 - 3m}{2}\right) = 15.5$$

$$14m + 19.5 - 9m = 31$$

$$5m = 11.5$$

$$m = 2.3$$

$$b = \frac{6.5 - 3(2.3)}{2} = -0.2$$

$$\hat{y} = mx + b = 2.3x - 0.2$$

(0, 0.5) $\hat{y} = 2.3(0) - 0.2 = -0.2$
 $e = \hat{y} - y = -0.2 - 0.5 = -0.7$
 $e^2 = (-0.7)^2 = 0.49$

(1, 1.5) $\hat{y} = 2.3(1) - 0.2 = 2.1$
 $e = \hat{y} - y = 2.1 - 1.5 = 0.6$
 $e^2 = (0.6)^2 = 0.36$

(2, 3.5) $\hat{y} = 2.3(2) - 0.2 = 4.4$
 $e = \hat{y} - y = 4.4 - 3.5 = 0.9$
 $e^2 = (0.9)^2 = 0.81$

(3, 7.5) $\hat{y} = 2.3(3) - 0.2 = 6.7$
 $e = \hat{y} - y = 6.7 - 7.5 = -0.8$
 $e^2 = (-0.8)^2 = 0.64$

$$MSE = \frac{1}{4} \sum_{i=1}^4 e^2 = \frac{1}{4} (0.49 + 0.81 + 0.36 + 0.64) = 0.575$$

— Newton-Raphson

$$MSE(m, b) = 3.5m^2 + b^2 + 3mb - 15.5m - 6.5b + 17.75$$

$$\frac{\partial MSE(m, b)}{\partial m} = 7.5m + 3b - 15.5$$

$$\frac{\partial^2 MSE(m, b)}{\partial m^2} = 7.5$$

$$\rightarrow m_{k+1} = m_k - \frac{7.5m_k + 3b_k - 15.5}{7.5} = \frac{15.5 - 3b_k}{7.5}$$

$$\frac{\partial MSE(m, b)}{\partial b} = 2b + 3m - 6.5$$

$$\frac{\partial^2 MSE(m, b)}{\partial b^2} = 2$$

$$\rightarrow b_{k+1} = b_k - \frac{2b_k + 3m_{k+1} - 6.5}{2} = \frac{6.5 - 3m_{k+1}}{2}$$

① $m_0 = 0$ $b_0 = 0$

$$m_1 = \frac{15.5 - 3(0)}{7.5} = 2.214286$$

$$b_1 = \frac{6.5 - 3(2.214286)}{2} = -0.071429$$

$$\hat{y}(0) = 2.214286(0) - 0.071429 = -0.071429$$

$$e^2 = (-0.071429 - 0.5)^2 = 0.326531$$

$$\hat{y}(2) = 2.214286(2) - 0.071429 = 4.357143$$

$$e^2 = (4.357143 - 3.5)^2 = 0.734694$$

$$\hat{y}(1) = 2.214286(1) - 0.071429 = 2.142857$$

$$e^2 = (2.142857 - 1.5)^2 = 0.413265$$

$$\hat{y}(3) = 2.214286(3) - 0.071429 = 6.571429$$

$$e^2 = (6.571429 - 7.5)^2 = 0.862245$$

$$MSE = \frac{0.326531 + 0.734694 + 0.413265 + 0.862245}{4} = 0.584184$$

$$\begin{aligned} \textcircled{2} \quad m_2 &= \frac{15.5 - 3(-0.071429)}{7} = 2.244898 \\ b_2 &= \frac{6.5 - 3(2.244898)}{2} = -0.117347 \\ \hat{y}(0) &= 2.244898(0) - 0.117347 = -0.117347 & e^2 &= (-0.117347 - 0.5)^2 = 0.381117 \\ \hat{y}(2) &= 2.244898(2) - 0.117347 = 4.372449 & e^2 &= (4.372449 - 3.5)^2 = 0.761167 \\ \hat{y}(1) &= 2.244898(1) - 0.117347 = 2.127551 & e^2 &= (2.127551 - 1.5)^2 = 0.393820 \\ \hat{y}(3) &= 2.244898(3) - 0.117347 = 6.617347 & e^2 &= (6.617347 - 7.5)^2 = 0.779076 \\ MSE &= \frac{0.381117 + 0.761167 + 0.393820 + 0.779076}{4} = 0.578795 \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad m_3 &= \frac{15.5 - 3(-0.117347)}{7} = 2.264577 \\ b_3 &= \frac{6.5 - 3(2.264577)}{2} = -0.146866 \\ \hat{y}(0) &= 2.264577(0) - 0.146866 = -0.146866 & e^2 &= (-0.146866 - 0.5)^2 = 0.418436 \\ \hat{y}(2) &= 2.264577(2) - 0.146866 = 4.382288 & e^2 &= (4.382288 - 3.5)^2 = 0.778432 \\ \hat{y}(1) &= 2.264577(1) - 0.146866 = 2.117711 & e^2 &= (2.117711 - 1.5)^2 = 0.381567 \\ \hat{y}(3) &= 2.264577(3) - 0.146866 = 6.646865 & e^2 &= (6.646865 - 7.5)^2 = 0.727839 \\ MSE &= \frac{0.418436 + 0.778432 + 0.381567 + 0.727839}{4} = 0.576568 \end{aligned}$$

— Parabola

— Analytical : $y = ax^2 + bx + c$

$$\hat{y} = ax_i^2 + bx_i + c \longrightarrow \text{error } e_i = \hat{y}_i - y_i$$

$$\begin{aligned} \text{For 4 points: } MSE(a, b, c) &= \frac{1}{4} \sum_{i=1}^4 (ax_i^2 + bx_i + c - y_i)^2 \\ &= \frac{1}{4} [(c - 0.5)^2 + (4a + 2b + c - 3.5)^2 + (a + b + c - 1.5)^2 + (9a + 3b + c - 7.5)^2] \\ &= \frac{1}{4} [c^2 - c + 0.25 + 16a^2 + 4b^2 + c^2 + 16ab + 8ac + 4bc - 28a - 14b - 7c + 12.25 \\ &\quad + a^2 + b^2 + c^2 + 2ab + 2ac + 2bc - 3a - 3b - 3c + 2.25 \\ &\quad + 81a^2 + 9b^2 + c^2 + 54ab + 18ac + 6bc - 135a - 45b - 15c + 56.25] \\ &= \frac{1}{4} (98a^2 + 14b^2 + 4c^2 + 72ab + 28ac + 12bc - 166a - 62b - 26c + 71) \\ &= 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5a - 15.5b - 6.5c + 17.75 \end{aligned}$$

$$\begin{aligned} \frac{\partial MSE(a, b, c)}{\partial a} &= 49a + 18b + 7c - 41.5 = 0 \longrightarrow \\ \frac{\partial MSE(a, b, c)}{\partial b} &= 7b + 18a + 3c - 15.5 = 0 \longrightarrow \\ \frac{\partial MSE(a, b, c)}{\partial c} &= 2c + 7a + 3b - 6.5 = 0 \longrightarrow \end{aligned}$$

$$\begin{cases} 49a + 18b + 7c = 41.5 \\ 18a + 7b + 3c = 15.5 \\ 7a + 3b + 2c = 6.5 \end{cases}$$

$$\hat{y} = 0.75x^2 + 0.05x + 0.55$$

$$\begin{aligned} (0, 0.5) \quad \hat{y} &= 0.75(0)^2 + 0.05(0) + 0.55 = 0.55 \\ e &= \hat{y} - y = 0.55 - 0.5 = 0.05 \\ e^2 &= (0.05)^2 = 0.0025 \\ (2, 3.5) \quad \hat{y} &= 0.75(2)^2 + 0.05(2) + 0.55 = 3.65 \\ e &= \hat{y} - y = 3.65 - 3.5 = 0.15 \\ e^2 &= (0.15)^2 = 0.0225 \end{aligned}$$

$$\begin{aligned} (1, 1.5) \quad \hat{y} &= 0.75(1)^2 + 0.05(1) + 0.55 = 1.35 \\ e &= \hat{y} - y = 1.35 - 1.5 = -0.15 \\ e^2 &= (-0.15)^2 = 0.0225 \\ (3, 7.5) \quad \hat{y} &= 0.75(3)^2 + 0.05(3) + 0.55 = 7.45 \\ e &= \hat{y} - y = 7.45 - 7.5 = -0.05 \\ e^2 &= (-0.05)^2 = 0.0025 \end{aligned}$$

$$MSE = \frac{1}{4} \sum_{i=1}^4 e^2 = \frac{1}{4} (0.0025 + 0.0225 + 0.0225 + 0.0025) = 0.0125$$

— Newton - Raphson

$$MSE(a,b,c) = 24.5a^2 + 3.5b^2 + c^2 + 18ab + 7ac + 3bc - 41.5 - 15.5b - 6.5c + 17.75$$

$$\left. \begin{aligned} \frac{\partial MSE(a,b,c)}{\partial a} &= 49a + 18b + 7c - 41.5 \\ \frac{\partial^2 MSE(a,b,c)}{\partial a^2} &= 49 \end{aligned} \right\} \longrightarrow a_{k+1} = a_k - \frac{49a_k + 18b_k + 7c_k - 41.5}{49} = \boxed{\frac{41.5 - 18b_k - 7c_k}{49}}$$

$$\left. \begin{aligned} \frac{\partial MSE(a,b,c)}{\partial b} &= 18a + 7b + 3c - 15.5 \\ \frac{\partial^2 MSE(a,b,c)}{\partial b^2} &= 7 \end{aligned} \right\} \longrightarrow b_{k+1} = b_k - \frac{18a_{k+1} + 7b_k + 3c_k - 15.5}{7} = \boxed{\frac{15.5 - 18a_{k+1} - 3c_k}{7}}$$

$$\left. \begin{aligned} \frac{\partial MSE(a,b,c)}{\partial c} &= 7a + 3b + 2c - 6.5 \\ \frac{\partial^2 MSE(a,b,c)}{\partial c^2} &= 2 \end{aligned} \right\} \longrightarrow c_{k+1} = c_k - \frac{7a_{k+1} + 3b_{k+1} + 2c_k - 6.5}{2} = \boxed{\frac{6.5 - 7a_{k+1} - 3b_{k+1}}{2}}$$

① $a_0 = 0 \quad b_0 = 0 \quad c_0 = 0$

$$a_1 = \frac{41.5 - 18(0) - 7(0)}{49} = \boxed{0.846939}$$

$$b_1 = \frac{15.5 - 18(0.846939) - 3(0)}{7} = \boxed{0.036443}$$

$$c_1 = \frac{6.5 - 7(0.846939) - 3(0.036443)}{2} = \boxed{0.231050}$$

$$\hat{y}(0) = 0.846939(0)^2 + 0.036443(0) + 0.231050 = 0.231050$$

$$\hat{y}(2) = 0.846939(2)^2 + 0.036443(2) + 0.231050 = 3.691692$$

$$\hat{y}(1) = 0.846939(1)^2 + 0.036443(1) + 0.231050 = 1.114432$$

$$\hat{y}(3) = 0.846939(3)^2 + 0.036443(3) + 0.231050 = 7.962830$$

$$MSE = \frac{0.072334 + 0.036746 + 0.148663 + 0.214212}{4} = \boxed{0.117989}$$

$$e^2 = (0.231050 - 0.5)^2 = 0.072334$$

$$e^2 = (3.691692 - 3.5)^2 = 0.036746$$

$$e^2 = (1.114432 - 1.5)^2 = 0.148663$$

$$e^2 = (7.962830 - 7.5)^2 = 0.214212$$

② $a_2 = \frac{41.5 - 18(0.036443) - 7(0.231050)}{49} = \boxed{0.800544}$

$$b_2 = \frac{15.5 - 18(0.800544) - 3(0.231050)}{7} = \boxed{0.056722}$$

$$c_2 = \frac{6.5 - 7(0.800544) - 3(0.056722)}{2} = \boxed{0.363012}$$

$$\hat{y}(0) = 0.800544(0)^2 + 0.056722(0) + 0.363012 = 0.363012$$

$$\hat{y}(2) = 0.800544(2)^2 + 0.056722(2) + 0.363012 = 3.678632$$

$$\hat{y}(1) = 0.800544(1)^2 + 0.056722(1) + 0.363012 = 1.220278$$

$$\hat{y}(3) = 0.800544(3)^2 + 0.056722(3) + 0.363012 = 7.738074$$

$$MSE = \frac{0.018766 + 0.031909 + 0.078244 + 0.056679}{4} = \boxed{0.046400}$$

$$e^2 = (0.363012 - 0.5)^2 = 0.018766$$

$$e^2 = (3.678632 - 3.5)^2 = 0.031909$$

$$e^2 = (1.220278 - 1.5)^2 = 0.078244$$

$$e^2 = (7.738074 - 7.5)^2 = 0.056679$$

③ $a_3 = \frac{41.5 - 18(0.056722) - 7(0.363012)}{49} = \boxed{0.774243}$

$$b_3 = \frac{15.5 - 18(0.774243) - 3(0.363012)}{7} = \boxed{0.067798}$$

$$c_3 = \frac{6.5 - 7(0.774243) - 3(0.067798)}{2} = \boxed{0.438452}$$

$$\hat{y}(0) = 0.774243(0)^2 + 0.067798(0) + 0.438452 = 0.438452$$

$$\hat{y}(2) = 0.774243(2)^2 + 0.067798(2) + 0.438452 = 3.671020$$

$$\hat{y}(1) = 0.774243(1)^2 + 0.067798(1) + 0.438452 = 1.280493$$

$$\hat{y}(3) = 0.774243(3)^2 + 0.067798(3) + 0.438452 = 7.610033$$

$$MSE = \frac{0.003786 + 0.029248 + 0.048183 + 0.012107}{4} = \boxed{0.02332}$$

$$e^2 = (0.438452 - 0.5)^2 = 0.003786$$

$$e^2 = (3.671020 - 3.5)^2 = 0.029248$$

$$e^2 = (1.280493 - 1.5)^2 = 0.048183$$

$$e^2 = (7.610033 - 7.5)^2 = 0.012107$$